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Lab Reports of Mathematics-II

Submitted by: -

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BCA 2nd Semester, 2024

Symbol Number:

Submitted to: -

.....

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DECLARATION

I hereby declare that the Mathematics Lab Report entitled “**Lab Reports of Mathematics-II**” submitted by me is an original and authentic work completed as a part of the academic requirements for the **Bachelor of Computer Applications (BCA)** program under **Tribhuvan University**.

All practical exercises, simulations, and analyses presented in this report were conducted using **MATLAB** and **Mathematica** software. Proper acknowledgment has been provided for all references, resources, and materials consulted during the preparation of this report.

I further confirm that this report has not been submitted, either in whole or in part, to fulfill the requirements of any other academic program. I have maintained strict adherence to the principles of academic integrity and honesty throughout the entire process of this work.

I would also like to express my sincere gratitude to my supervisor, **Mr. Harka Bahadur Shrestha**, for his invaluable guidance, encouragement, and support throughout the completion of this lab work.

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Date:

CERTIFICATE FROM THE SUPERVISOR

This is to certify that the laboratory report titled "**Lab Reports of Mathematics-II**" has been diligently prepared by **Bishesh Upreti** in fulfillment of the academic requirements for the **Bachelor of Computer Applications (BCA)** degree at the Faculty of Humanities and Social Sciences, Tribhuvan University.

Throughout the completion of this report, Samjhauta Dahal has demonstrated a commendable understanding of mathematical concepts and proficient utilization of **MATLAB** and **Mathematica** software. The report showcases Bishesh Upreti's dedication to academic excellence and their adeptness in applying practical skills and knowledge gained during lab sessions.

I hereby verify that, to the best of my knowledge, the work presented in this lab report is solely the original creation of **Bishesh upreti**.

Harka Bahadur Shrestha

Internal Supervisor
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Date: 21st August 2025

Table of Contents

DECLARATION	2
CERTIFICATE FROM THE SUPERVISOR	3
MATHEMATICA	5
Introduction to MATHEMATICA.....	5
Lab 1. Limits of Function	6
Lab2. Derivatives of Function	8
Lab 3. Maximization and Minimization of function	10
Lab 4. Graph of Function.....	11
Lab 5. Integration	12
MATLAB	14
Introduction to MATLAB	14
Lab 1. Matrix Inversion Method	15
Lab 2. Linear Programming Problem	16
Lab 3. Bisection Method	18
Lab 5. Trapezoidal Rule	19
Lab 6. Simpson's 1/3 Rule	20
Used Software.....	21
References	21

MATHEMATICA

Introduction to MATHEMATICA

Mathematica is a powerful computational software system developed by **Wolfram Research**, first released in 1988 and led by **Stephen Wolfram**. It is widely used in scientific, engineering, mathematical, and computing fields for its ability to perform symbolic and numerical computations, visualize data, and automate tasks.

The core strength of Mathematica lies in its **Wolfram Language**, a highly symbolic and functional programming language designed for computational mathematics and knowledge-based programming. The language supports a vast range of operations, including algebraic manipulations, calculus, matrix operations, data analysis, plotting, and even machine learning.

Key Features Of MATHEMATICA

- Symbolic and Numeric Computation
- High-Quality Visualization and Graphics
- Built-in Knowledgebase (Wolfram Knowledgebase)
- Interactive Notebooks
- Support for Machine Learning and Neural Networks
- Advanced Data Import/Export Capabilities
- Cloud Integration via Wolfram Cloud

Mathematica is commonly used in academia, research, and industry, offering a unified platform for both prototyping and deployment of complex algorithms.

Lab 1. Limits of Function

A **limit** describes the value that a function (or sequence) approaches as the input (or index) approaches a particular value.

Mathematically,

$L = \lim_{x \rightarrow a} f(x)$, As x gets closer and closer to a , the function $f(x)$ gets closer and closer to L .

Q1. Solve Following Algebraic limits problem Using Mathematica.

a. $\lim_{x \rightarrow 5} \frac{4}{x-7}$

```
In[22]:= Limit[4 / (x - 7), x -> 5]  
Out[22]= -2
```

b. $\lim_{x \rightarrow 2} \frac{x^2-25}{x-5}$

```
In[2]:= Limit[(x^2 - 25) / (x - 5), x -> 2]  
Out[2]= 7
```

c. $\lim_{x \rightarrow -3} \frac{x+3}{x^2+4x+3}$

```
In[3]:= Limit[(x + 3) / (x^2 + 4 x + 3), x -> -3]  
Out[3]= -1/2
```

d. $\lim_{x \rightarrow -1} \frac{\sqrt{x^2+8}-3}{x+1}$

```
In[6]:= Limit[((x^2 + 8)^(1/2) - 3) / (x + 1), x -> -1]  
Out[6]= -1/3
```

e. $\lim_{x \rightarrow \infty} \sqrt{x+1} - \sqrt{x}$

```
In[7]:= Limit[(x + 1)^(1/2) - x^(1/2), x -> Infinity]  
Out[7]= 0
```

Q2. Solve Following Trigonometric limits problem Using Mathematica.

a. $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{1 - \cos 5x}$

```
In[8]:= Limit[(1 - Cos[3 x]) / (1 - Cos[5 x]), x -> 0]  
Out[8]= 9/25
```

b. $\lim_{x \rightarrow \theta} \frac{x \sin \theta - \theta \sin x}{x - \theta}$

```
In[9]:= Limit[(x Sin[θ] - θ Sin[x]) / (x - θ), x -> θ]  
Out[9]= -θ Cos[θ] + Sin[θ]
```

c. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{x - \frac{\pi}{4}}$

```
In[10]:= Limit[(Sin[x] - Cos[x]) / (x - (Pi / 4)), x -> (Pi / 4)]
Out[10]=  $\sqrt{2}$ 
```

Q1. Solve Following Exponential and Logarithmic limits problem Using Mathematica.

a. $\lim_{x \rightarrow 0} \frac{e^x - 1}{3x}$

```
In[1]:= Limit[(E^(x) - 1) / (3 x), x -> 0]
Out[1]=  $\frac{1}{3}$ 
```

b. $\lim_{x \rightarrow 0} \frac{\log_e(1+5x)}{2x}$

```
In[4]:= Limit[(Log[E, 1 + 5 x]) / (2 x), x -> 0]
Out[4]=  $\frac{5}{2}$ 
```

c. $\lim_{x \rightarrow 0} \frac{3^{2x} - 2^{3x}}{x}$

```
In[5]:= Limit[(3^(2 x) - 2^(3 x)) / x, x -> 0]
Out[5]=  $\text{Log}\left[\frac{9}{8}\right]$ 
```

Lab2. Derivatives of Function

The Derivatives of a function measure rate at which the function's value change with respect to change in its input (usually denoted as x)

Derivatives gives the slope of the tangent line to curve at a particular point.

Mathematically,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}, \text{ Where } h \text{ be the small change in on } x.$$

Q1. Find the Derivatives of following Using Mathematica.

a. $f(x) = \frac{ax+b}{x}$

```
In[12]:= D[(2 x + b) / x, x] // Simplify  
Out[12]=  $-\frac{b}{x^2}$ 
```

b. $f(x) = \frac{1}{\sqrt{x}}$

```
In[21]:= D[1 / x ^ (1 / 2), x]  
Out[21]=  $-\frac{1}{2 x^{3/2}}$ 
```

c. $f(x) = \sqrt{\sin 2x}$

```
In[22]:= D[Sqrt[Sin[2 x]], x]  
Out[22]=  $\frac{\text{Cos}[2 x]}{\sqrt{\text{Sin}[2 x]}}$ 
```

d. $f(x) = \tan^2 \sin(\sqrt{x})$

```
In[24]:= D[(Tan[Sin[Sqrt[x]]])^2, x]  
Out[24]=  $\frac{\text{Cos}[\sqrt{x}] \text{Sec}[\text{Sin}[\sqrt{x}]]^2 \text{Tan}[\text{Sin}[\sqrt{x}]]}{\sqrt{x}}$ 
```

e. $f(x) = \cos^{-1} x^2$

```
In[25]:= D[ArcCos[x ^ 2], x]  
Out[25]=  $-\frac{2 x}{\sqrt{1 - x^4}}$ 
```

f. $f(x) = e^{2x+3}$

```
In[26]:= D[E ^ (2 x + 3), x]  
Out[26]=  $2 e^{3+2 x}$ 
```


g. $f(x) = \ln(3x + 5)$

```
In[27]:= D[Log[E, 3 x + 5], x]
Out[27]=  $\frac{3}{5 + 3 x}$ 
```

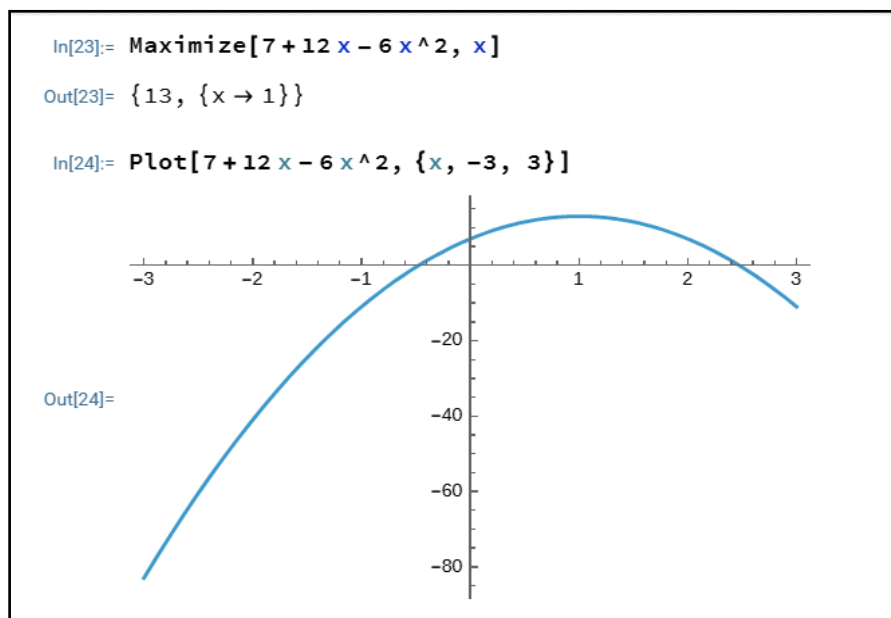
h. $f(x) = e^{\sin x}$

```
In[28]:= D[E ^ Sin[x], x]
Out[28]=  $e^{\sin[x]} \cos[x]$ 
```

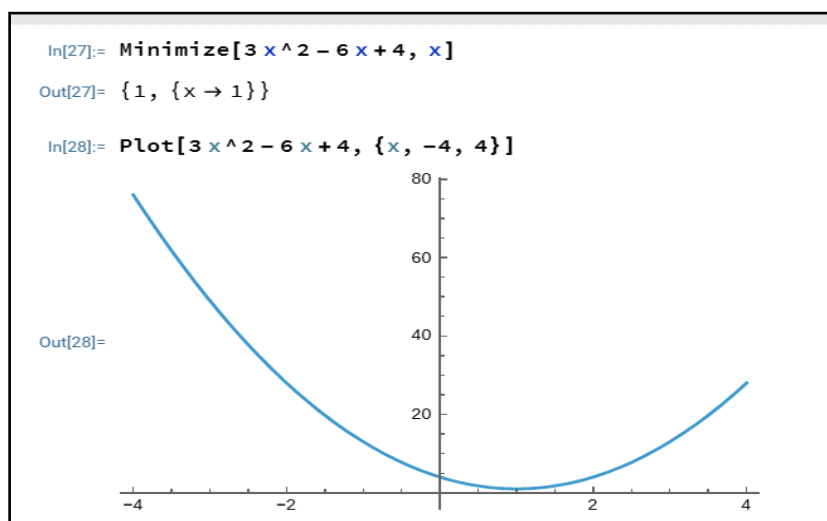
Lab 3. Maximization and Minimization of function

Maximization is the process of finding the highest possible value of a function within a specified domain, while minimization seeks the lowest possible value, both achieved by identifying critical points (where the derivative $f'(x) = 0$) and confirming peaks (maxima) or valleys (minima) using derivative tests. These techniques optimize real-world objectives like maximizing profit or minimizing cost.

1. Find the maximum value of $f(x) = -6x^2 + 12x + 7$



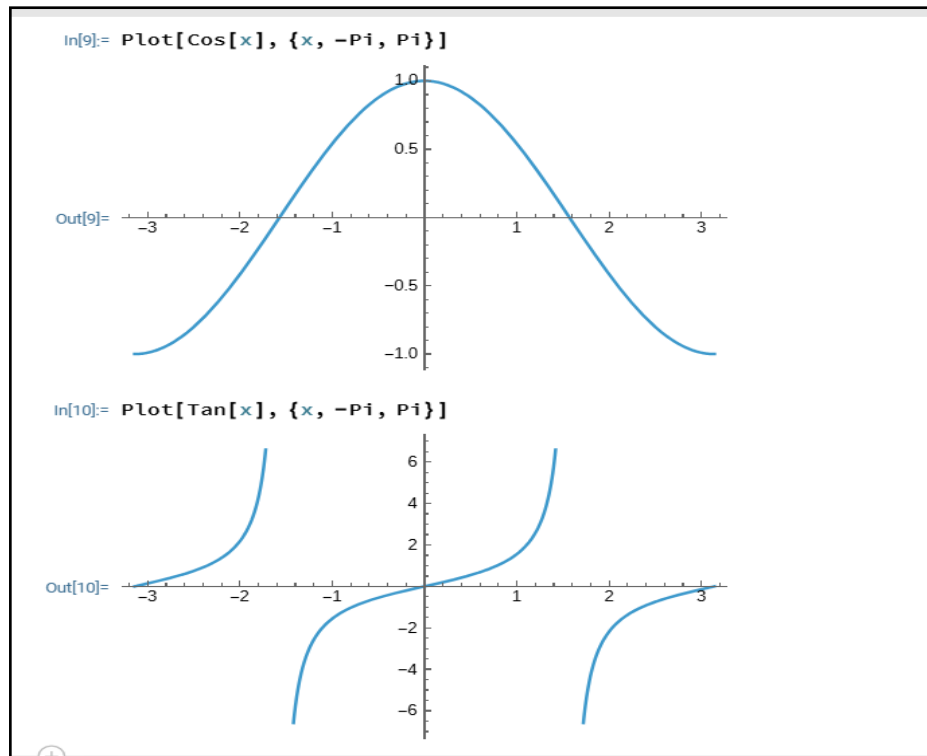
2. Find the Minimum value of $f(x) = 3x^2 - 6x + 4$



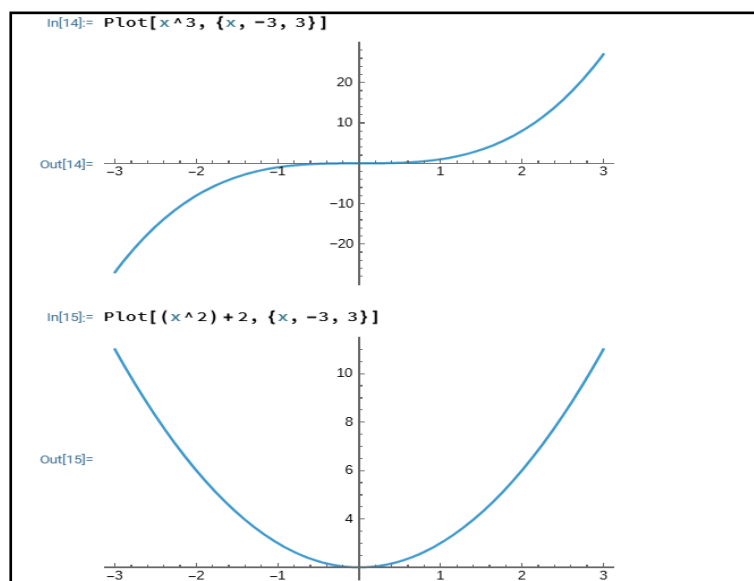
Lab 4. Graph of Function.

In 'Mathematics II' we learn to graph algebraic functions using first and second order derivatives. However, in Mathematica we can easily graph any function.

1. Graph of $\cos x$ and $\tan x$



2. Graph of $y = x^3$ and $y = x^2 + 2$



Lab 5. Integration

Integration is the mathematical operation that computes the accumulation of quantities—such as areas under curves, total displacement from velocity, or aggregated change—by summing infinitesimal contributions. It reverses differentiation (finding antiderivatives: if $F'(x) = f(x)$ then $\int f(x) dx = F(x) + C$ and is formally defined via limits of sums (e.g., Riemann sums). Integrals are categorized as indefinite (general form $\int f(x)dx$, yielding a family of functions) or definite (specific value $\int_a^b f(x)dx$, representing net accumulation between bounds a and b)

1. Indefinite Integral

Evaluate following

a. $\int (x + 1) dx$

```
In[1]:= Integrate[(x + 1), x]
```

```
Out[1]= x +  $\frac{x^2}{2}$ 
```

b. $\int \sin 4x \cos 2x dx$

```
In[9]:= Integrate[Sin[4 x] * Cos[2 x], x]
```

```
Out[9]=  $-\frac{1}{2} \cos[x]^2 - \frac{1}{12} \cos[6 x]$ 
```

c. $\int \frac{\sin \log x}{x} dx$

```
In[10]:= Integrate[Sin[Log[x]] / x, x]
```

```
Out[10]= -Cos[Log[x]]
```

d. $\int e^x \cos x dx$

```
In[11]:= Integrate[E^x * Cos[x], x]
```

```
Out[11]=  $\frac{1}{2} e^x (\cos[x] + \sin[x])$ 
```

e. $\int \frac{2x-11}{x^2+x-2} dx$

```
In[12]:= Integrate[(2 x - 11) / (x^2 + x - 2), x]
```

```
Out[12]= -3 Log[1 - x] + 5 Log[2 + x]
```

2. Definite Integral

Evaluate following

a. $\int_0^1 \frac{16x}{8x^2+2} dx$

```
In[13]:= Integrate[(16 x) / (8 x^2 + 2), {x, 0, 1}]
```

```
Out[13]= Log[5]
```

b. $\int_0^{\frac{\pi}{2}} x \sin x dx$

```
In[14]:= Integrate[Sin[x], {x, 0, Pi/2}]
```

```
Out[14]= 1
```

c. $\int_0^{\pi} (1 + \cos x) dx$

```
In[15]:= Integrate[1 + Cos[x], {x, 0, Pi}]
Out[15]=  $\pi$ 
```

3. Improper Integral Evaluate following

a. $\int_0^{\infty} \frac{dx}{x}$

```
In[16]:= Integrate[1/x, {x, 0, Infinity}]
... Integrate: Integral of  $\frac{1}{x}$  does not converge on {0,  $\infty$ }.
Out[16]=  $\int_0^{\infty} \frac{1}{x} dx$ 
```

b. $\int_0^1 \frac{dx}{\sqrt{x}}$

```
In[20]:= Integrate[1/Sqrt[x], {x, 0, 1}]
Out[20]= 2
```

c. $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$

```
In[21]:= Integrate[1/(1+x^2), {x, -Infinity, Infinity}]
Out[21]=  $\pi$ 
```

MATLAB

Introduction to MATLAB

MATLAB (short for MATrix LABoratory) is a high-level programming language and environment developed by MathWorks. It is widely used for numerical computation, data analysis, algorithm development, modeling, simulation, and visualization.

Originally developed in the late 1970s by Cleve Moler, MATLAB has become a standard tool in both academic and industrial fields. Its matrix-based structure makes it especially useful for solving mathematical and engineering problems efficiently.

Key Features of MATLAB

- Matrix-based structure – MATLAB is fundamentally built around matrix and array operations, making it ideal for linear algebra.
- High-level programming language – It supports functions, loops, conditionals, and object-oriented programming.
- Extensive library of built-in functions – Provides thousands of ready-to-use tools for statistics, machine learning, control systems, image processing, and more.
- Advanced graphics and visualization – Offers 2D and 3D plotting, animations, and interactive tools for data representation.
- Integration with Simulink – Works seamlessly with Simulink for modeling and simulating dynamic systems.
- Interfacing capabilities – Allows integration with C/C++, Java, Python, and hardware platforms like Arduino and Raspberry Pi.

Applications of MATLAB

- Engineering – Used in areas such as signal processing, control system design, robotics, and embedded systems.
- Academics – A powerful tool for teaching, learning, and conducting mathematical and technical research.
- Finance – Applied in financial modeling, risk analysis, data analysis, and optimization.
- Biotech and Pharmaceuticals – Supports data analysis, modeling of biological systems, and simulation in drug development.
- Automotive Industry – Assists in the simulation of vehicle systems, development of control systems, and autonomous driving technologies.
- Aerospace – Used for flight simulation, control system design, and sensor fusion applications.

Lab 1. Matrix Inversion Method

Theory

The Matrix Inversion Method is a technique used to solve a system of linear equations of the form $AX = B$, where A is the coefficient matrix, X is the column vector of unknowns, and B is the constant vector. This method involves finding the inverse of matrix A (denoted as A^{-1}) and then multiplying it with B to get the solution vector X. The formula used is:

$$X = A^{-1}B$$

In MATLAB, we can easily implement this method by defining matrices A and B, calculating the inverse of A using the `inv()` function, and multiplying it with B. However, this method should only be used if matrix A is square and non-singular (i.e., has a non-zero determinant).

Practical Implementation

1. Given System of linear equations

$$x - 2y - z = 1$$

$$x - y + 2z = 9$$

$$2x - 3y - z = 4$$

The matrix form of given system.

$$\begin{bmatrix} 1 & -2 & -1 \\ 1 & -1 & 2 \\ 2 & -3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 4 \end{bmatrix}$$

$$\text{i.e. } AX = B \Rightarrow X = A^{-1}B$$

...(i)

Command Window

```
>> A = [1, -2, -1; 1, -1, 2; 2, -3, -1];  
>> B = [1; 9; 4];  
>> X = inv(A)*B;  
>> fprintf('x = %.2f, y = %.2f, z = %.2f', X(1), X(2), X(3));  
x = 2.00, y = -1.00, z = 3.00  
>>
```

$\therefore x = 2, y = -1 \text{ and } z = 3$

Lab 2. Linear Programming Problem

Theory

A **Linear Programming Problem (LPP)** is a mathematical method used to determine the best possible outcome (such as maximum profit or minimum cost) in a given mathematical model. It involves:

- **Objective Function:** A linear function to be maximized or minimized.
- **Constraints:** A set of linear inequalities or equations that restrict the values of the decision variables.
- **Non-Negativity Restriction:** Decision variables must be zero or positive.

The **Simplex Method** is a popular algorithm to solve LPPs that involve two or more variables. It works by moving from one vertex (corner point) of the feasible region to another in such a way that the objective function improves or remains the same.

Practical Implementation

- Using Simplex Method find optimal solution.

1. *Max. $Z = 50x + 80y$ subject to*

$$x + 2y \leq 32$$

$$3x + 4y \leq 84$$

$$x, y \geq 0$$

2. *Min. $C = 4x + 2y$ subject to*

$$x + y \geq 21$$

$$3x + y \geq 27$$

$$x, y \geq 0$$

Command Window

```
>> f = [-50, -80];  
>> A = [1, 2; 3, 4];  
>> b = [32; 84];  
>> lb = [0; 0];  
>> [x, Z] = linprog(f, A, b, [], [], lb);  
  
Optimal solution found.  
  
>> fprintf('x = %.2f, y = %.2f, Max. Z = %.2f\n', x(1), x(2), -Z);  
x = 20.00, y = 6, Max. Z = 1480.00  
>>
```

∴ *Max $Z = 1480$ at $x = 20$ and $y = 6$*

Command Window

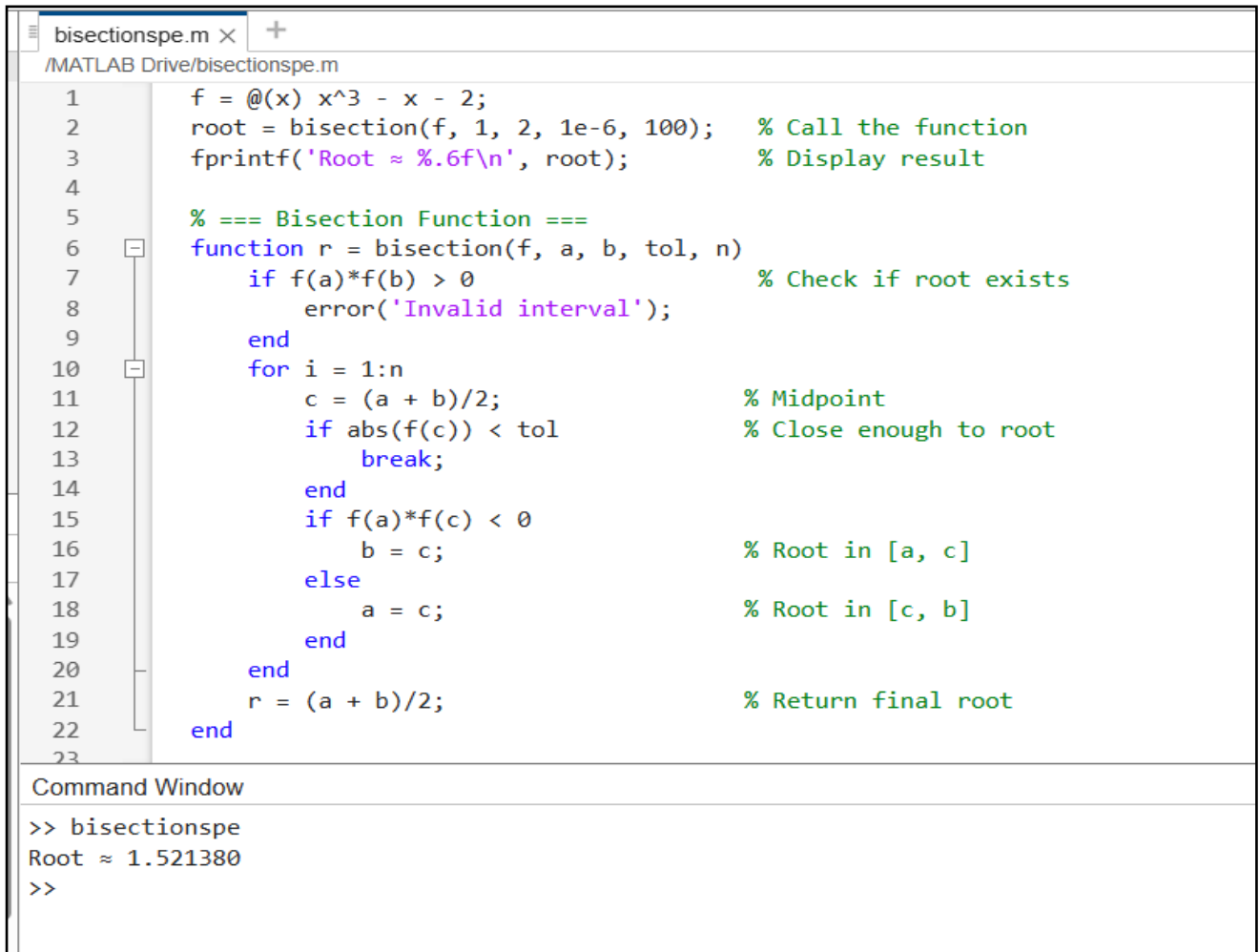
```
>> f=[4,2];  
>> A=[-1,-1;-3,-1];  
>> b=[-21;-27];  
>> lb=[0,0];  
>> [x, C] = linprog(f, A, b, [], [], lb);  
  
Optimal solution found.  
  
>> fprintf('x=%.2f, y=%.2f, Min. C =%.2f', x(1), x(2), C);  
x=3.00, y=18.00, Min. C =48.00  
>>  
>>
```

$\therefore$ Min. $C = 48$ at $x = 3$ and $y = 18$

Lab 3. Bisection Method

Bisection Method is a numerical technique to find the root of a function. It works by repeatedly halving an interval $[a, b]$ where the function changes sign (i.e., $f(a) \cdot f(b) < 0$) and narrowing down to the root. It is simple, reliable, and always converges for continuous functions.

Q. Use the Bisection Method in MATLAB to find an approximate root of the equation: $f(x) = x^3 - x + 2$ within the interval $[1, 2]$ using a tolerance of 10^{-6} and a maximum of 100 iterations.



```
bisectionspe.m × +
/MATLAB Drive/bisectionspe.m
1      f = @(x) x^3 - x - 2;
2      root = bisection(f, 1, 2, 1e-6, 100); % Call the function
3      fprintf('Root ≈ %.6f\n', root);      % Display result
4
5      % === Bisection Function ===
6      function r = bisection(f, a, b, tol, n)
7          if f(a)*f(b) > 0 % Check if root exists
8              error('Invalid interval');
9          end
10         for i = 1:n
11             c = (a + b)/2; % Midpoint
12             if abs(f(c)) < tol % Close enough to root
13                 break;
14             end
15             if f(a)*f(c) < 0
16                 b = c; % Root in [a, c]
17             else
18                 a = c; % Root in [c, b]
19             end
20         end
21         r = (a + b)/2; % Return final root
22     end
23
Command Window
>> bisectionspe
Root ≈ 1.521380
>>
```

Lab 5. Trapezoidal Rule

The Trapezoidal rule is a Numerical Method used to approximate the definite integral of function. It works by dividing the area under a curve into trapezoids rather than rectangles.

- **Formula:**

$$\int_a^b f(x) dx \approx \frac{h}{2} \left[f(a) + \sum_{i=1, 2, \dots}^{n-1} f(x_i) + f(b) \right]$$

Where,

n = total number of sub intervals,

$$h = \frac{b-a}{n},$$

$$x_i = a + h * i$$

Q. Using Composite Trapezoidal rule, Evaluate $\int_0^{\pi} \sin x \, dx$, n = 4

Command Window

```
>> f = @(x)sin(x);  
>> a = 0;  
>> b = pi;  
>> n = 4;  
>> h = (b-a)/n;  
>> s = 0.5*(f(a) + f(b));  
>> for i = 1:n-1  
s=s+f(a+i*h);  
end  
>> I = h*s
```

I =

1.8961

Lab 6. Simpson's 1/3 Rule

It is a method in numerical integration to approximate the definite integral of a function. It gives better approximation than the Trapezoidal Rule by using parabolas (quadratic Curves) instead of straight line.

- **Formula:**

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(a) + 4 \sum_{i=1, 3, \dots}^{n-1} f(x_i) + 2 \sum_{i=2, 4, \dots}^{n-2} f(x_i) + f(b) \right]$$

Where,

n = total number of sub intervals,

$$h = \frac{b-a}{n},$$

$$x_i = a + h * i$$

Q. Using Simpson's $\frac{1}{3}$ rule, Evaluate $\int_0^{\pi} \sin x \, dx$, n = 4

Command Window

```
>> f = @(x) sin(x);  
>> a = 0;  
>> b = pi;  
>> n = 4;  
>> h = (b-a)/n;  
>> s = f(a) + f(b);  
>> for i = 1:n-1  
    s = s + 4*f(a+i*h);  
end  
>> for k = 2:2:n-1;  
    s = s - 2*f(a+k*h);  
end  
>> I = (h/3)*s
```

I =

2.5282

Used Software

- **MATLAB** (Version: R2024b)
- **Wolfram Mathematica** (Version: 14.2.1)

References

1. **Jhavi Lal Ghimire** (2020). **Mathematics II**
Asmita Publication ISBN:978-9937-687-87-4.